

SQL & MySQL Assignment: Lab-0

1. What is SQL and Its Use?

SQL, or Structured Query Language, is the standard language used to communicate with relational database systems. It allows users to store, manipulate, and retrieve data efficiently. The primary purpose of SQL is to provide a consistent way to manage large sets of structured information.

Its uses include:

1. Data Retrieval: Extracting specific information from large datasets using queries.
2. Data Manipulation: Adding new records, updating existing ones, or deleting unnecessary data.
3. Database Definition: Creating and modifying tables, schemas, and other structures.
4. Access Control: Granting or revoking permissions to ensure data security.
5. Transaction Management: Ensuring data consistency and reliability when multiple operations are performed.

2. Describe all the different commands of SQL, mention use of it and syntax.

1. Data Definition Language (DDL)

- CREATE: Creates a new database or table.

Syntax: CREATE TABLE table_name (column1 datatype, column2 datatype);

- ALTER: Modifies an existing database object (adds, deletes, or modifies columns).

Syntax: ALTER TABLE table_name ADD column_name datatype;

- DROP: Deletes an entire table, view, or database permanently.

Syntax: DROP TABLE table_name;

- **TRUNCATE:** Removes all records from a table, but keeps the table structure.

Syntax: `TRUNCATE TABLE table_name;`

- **RENAME:** Changes the name of an existing database object.

Syntax: `ALTER TABLE old_table_name RENAME TO new_table_name;`

2. Data Manipulation Language (DML)

- **INSERT:** Adds new rows of data into a table.

Syntax: `INSERT INTO table_name (col1, col2) VALUES (val1, val2);`

- **UPDATE:** Modifies existing data within a table.

Syntax: `UPDATE table_name SET col1 = val1 WHERE condition;`

- **DELETE:** Removes specific records from a table based on a condition.

Syntax: `DELETE FROM table_name WHERE condition;`

3. Data Query Language (DQL)

- **SELECT:** Retrieves data from one or more tables.

Syntax: `SELECT column1, column2 FROM table_name WHERE condition;`

4. Data Control Language (DCL)

- **GRANT:** Gives permissions to a user to access the database or perform specific tasks.

Syntax: `GRANT SELECT, INSERT ON table_name TO user_name;`

- **REVOKE:** Withdraws the access permissions previously given to a user.

Syntax: REVOKE SELECT, INSERT ON table_name FROM user_name;

5. Transaction Control Language (TCL)

- COMMIT: Permanently saves all changes made during the current transaction.

Syntax: COMMIT;

- ROLLBACK: Undoes changes made in the current transaction if an error occurs.

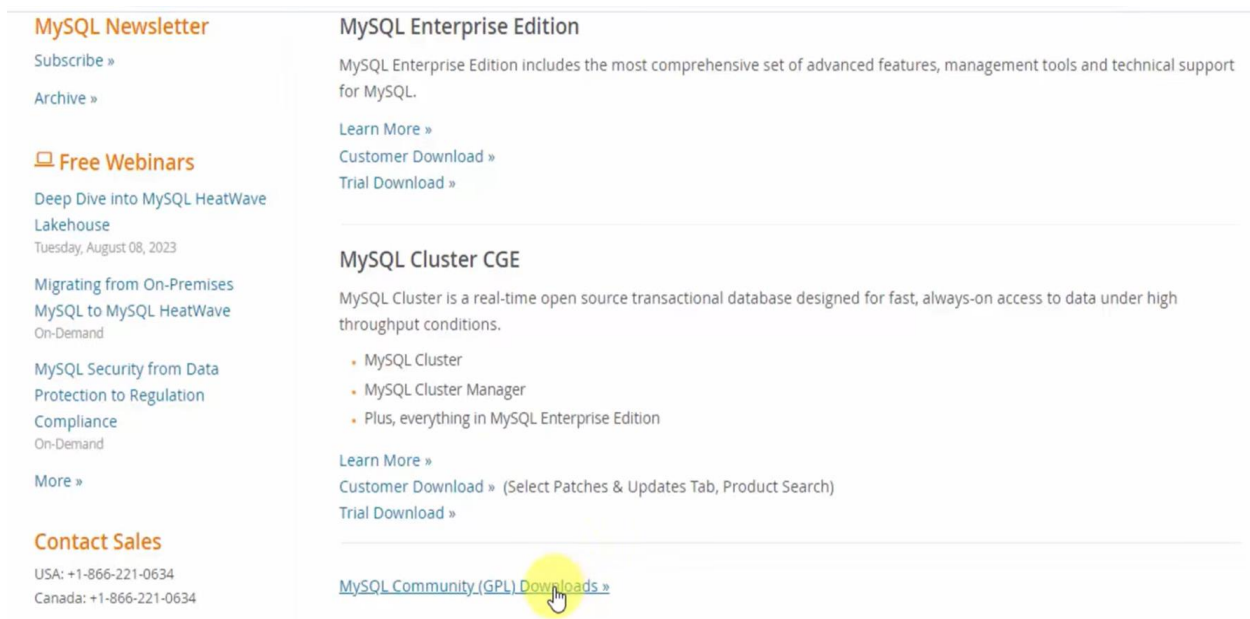
Syntax: ROLLBACK;

- SAVEPOINT: Sets a point within a transaction to which you can later roll back.

Syntax: SAVEPOINT savepoint_name

3. Write step by step procedure to install MYSQL.

Step 1: Go to the official MySQL Community Downloads page and select the latest MySQL Installer version for Windows.



The screenshot displays the MySQL website's navigation and product sections. On the left, there are links for the MySQL Newsletter, Free Webinars (including 'Deep Dive into MySQL HeatWave Lakehouse'), and Contact Sales. The main content area features 'MySQL Enterprise Edition' and 'MySQL Cluster CGE'. At the bottom right, a link for 'MySQL Community (GPL) Downloads' is highlighted with a yellow circle and a mouse cursor icon.

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More »

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[MySQL Community \(GPL\) Downloads »](#)

Step 2: Choose the installer type. You can either download the smaller web installer (which fetches components during setup) or the full installer (which contains all components).

General Availability (GA) Releases Archives

MySQL Installer 8.0.35

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:
8.0.35

Select Operating System:
Microsoft Windows

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.35.0.msi)	8.0.35	2.1M	Download
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.35.0.msi)	8.0.35	288.6M	Download

MD5: 214df2ccdf83eb5edc6ca7c115792406 | [Signature](#)

MD5: 2cfd4a448a2971b6b5323775ef9e8d012 | [Signature](#)

Step 3: When prompted, you may log in or sign up for an Oracle account. If you prefer, click on “*No thanks, just start my download*” to continue without an account

MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

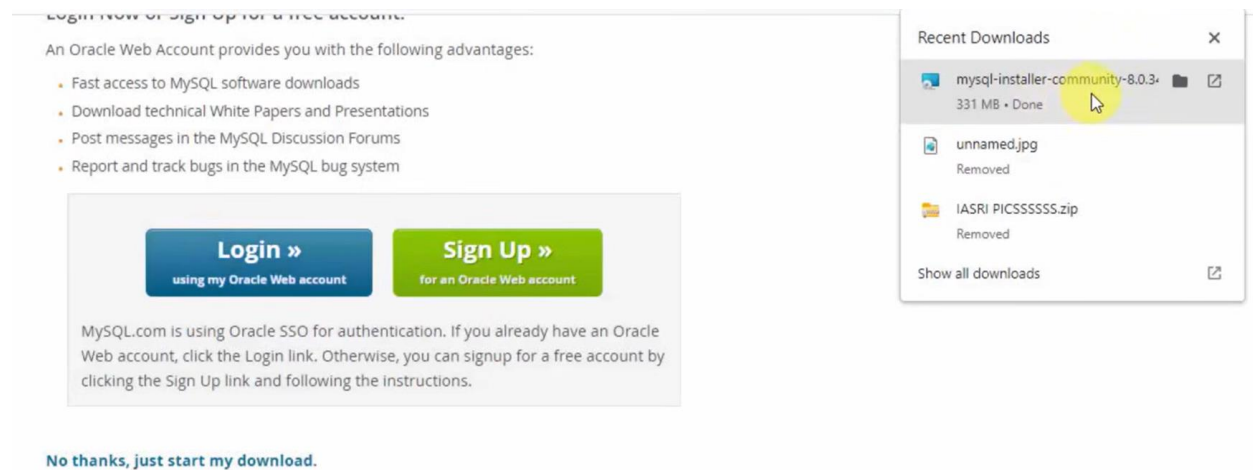
[Login »](#)
using my Oracle Web account

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for an Oracle Web account

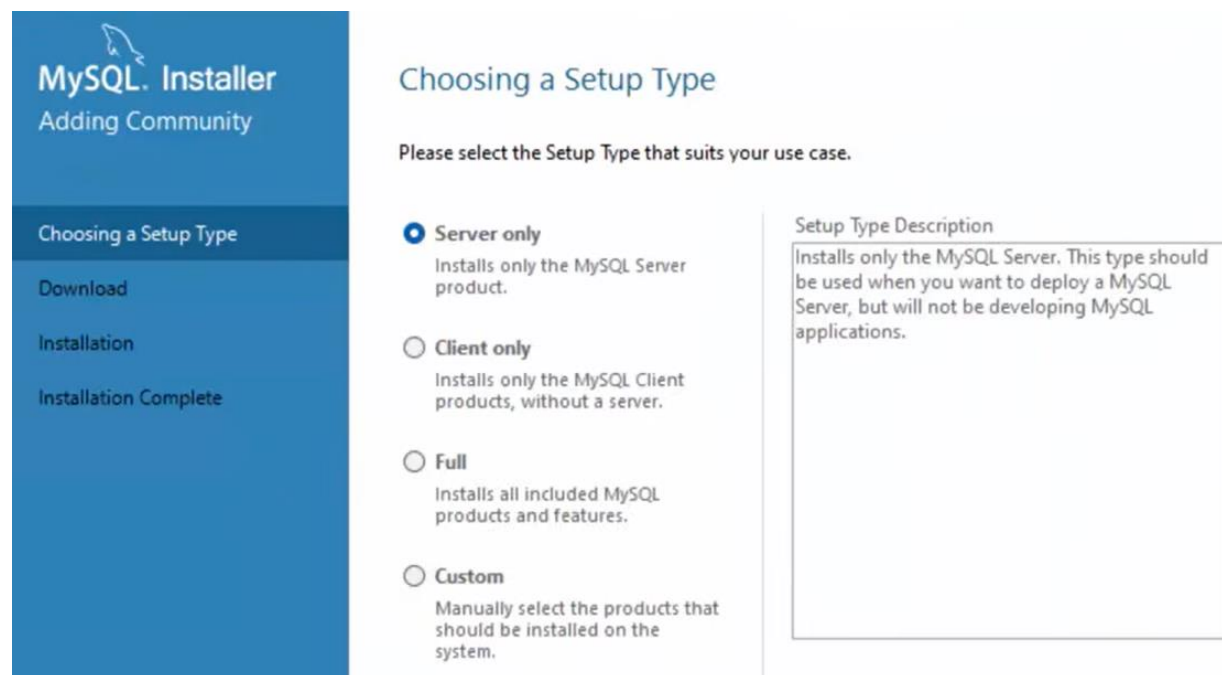
MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

[No thanks, just start my download.](#)

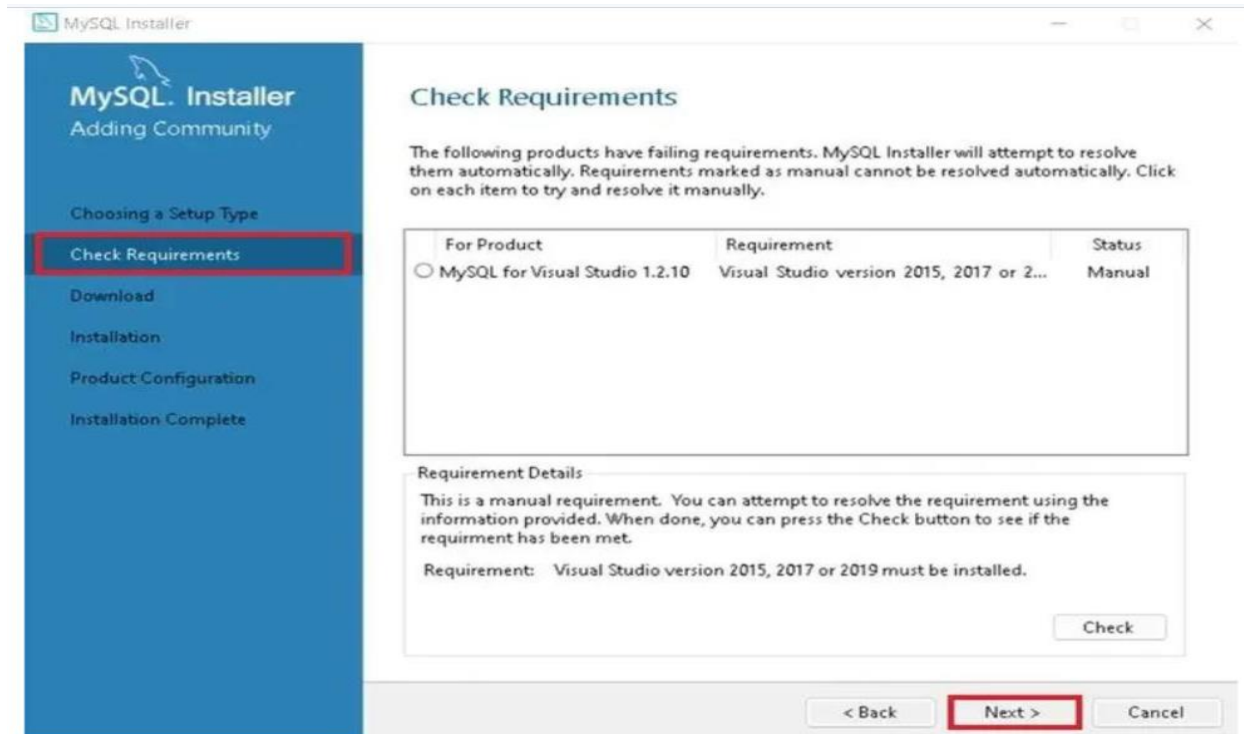
Step 4: Once downloaded, run the .msi installer file to launch the MySQL Installer wizard.



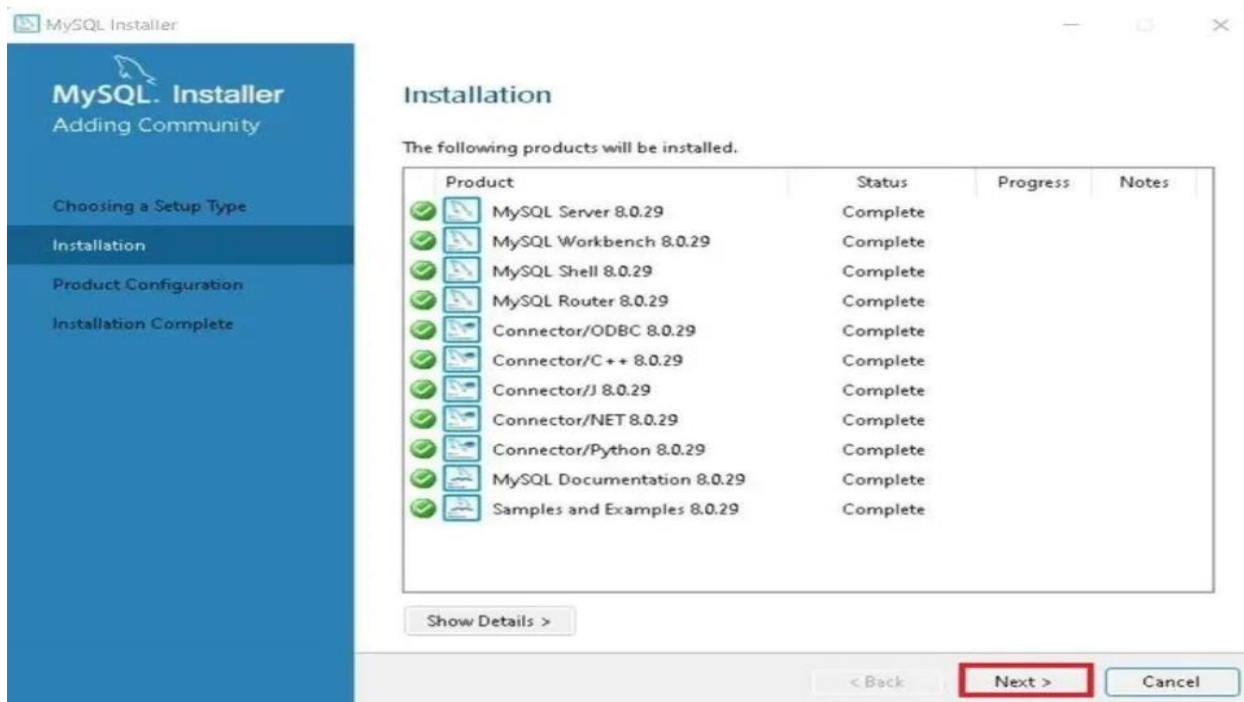
Step 5: Select the setup type. Options include Developer Default, Server Only, Client Only, or Custom. Choose the one that suits your requirement.



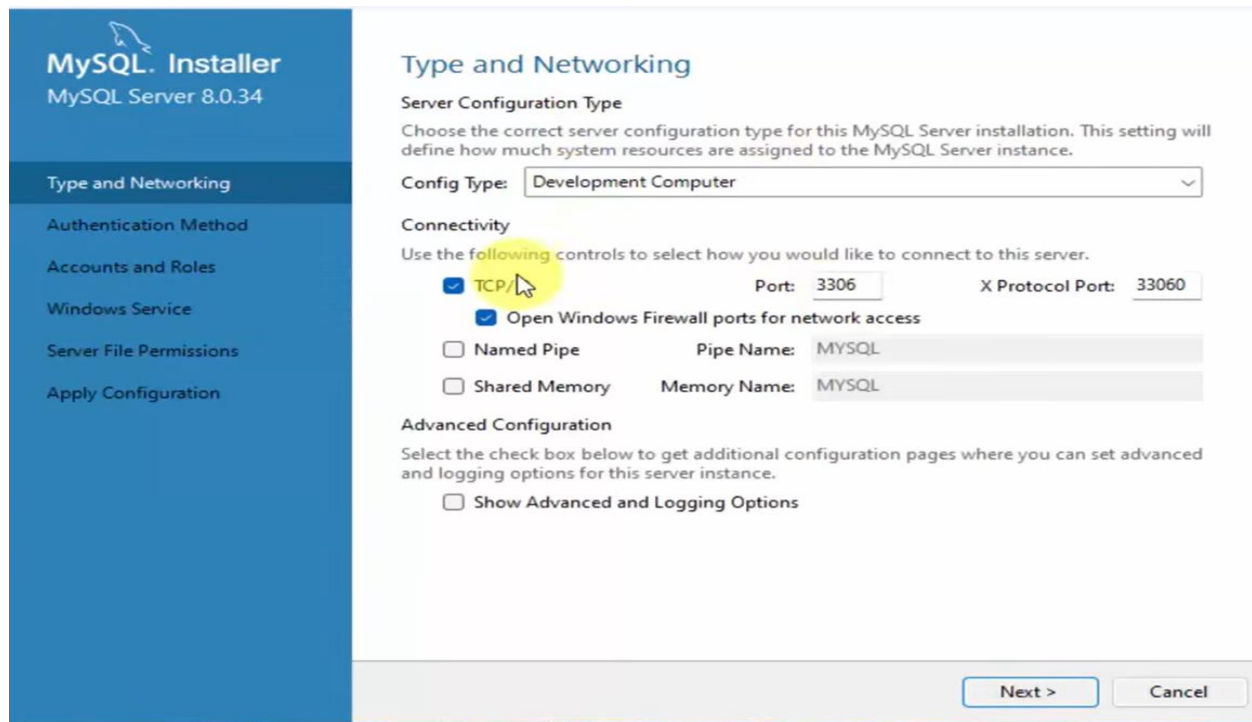
Step 6: The installer will check system requirements. If any prerequisites are missing (such as Visual Studio for MySQL for Visual Studio integration), resolve them before proceeding.



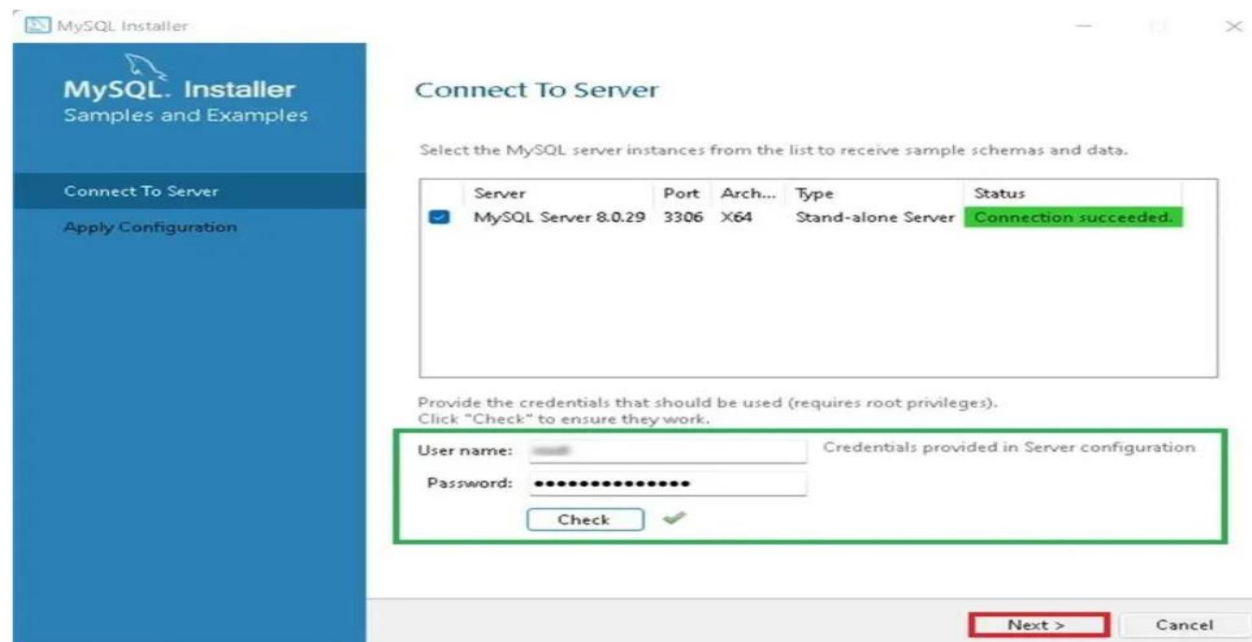
Step 7: The installer will now download and install the selected MySQL products such as MySQL Server, Workbench, Shell, Router, Connectors, Documentation, and Samples.



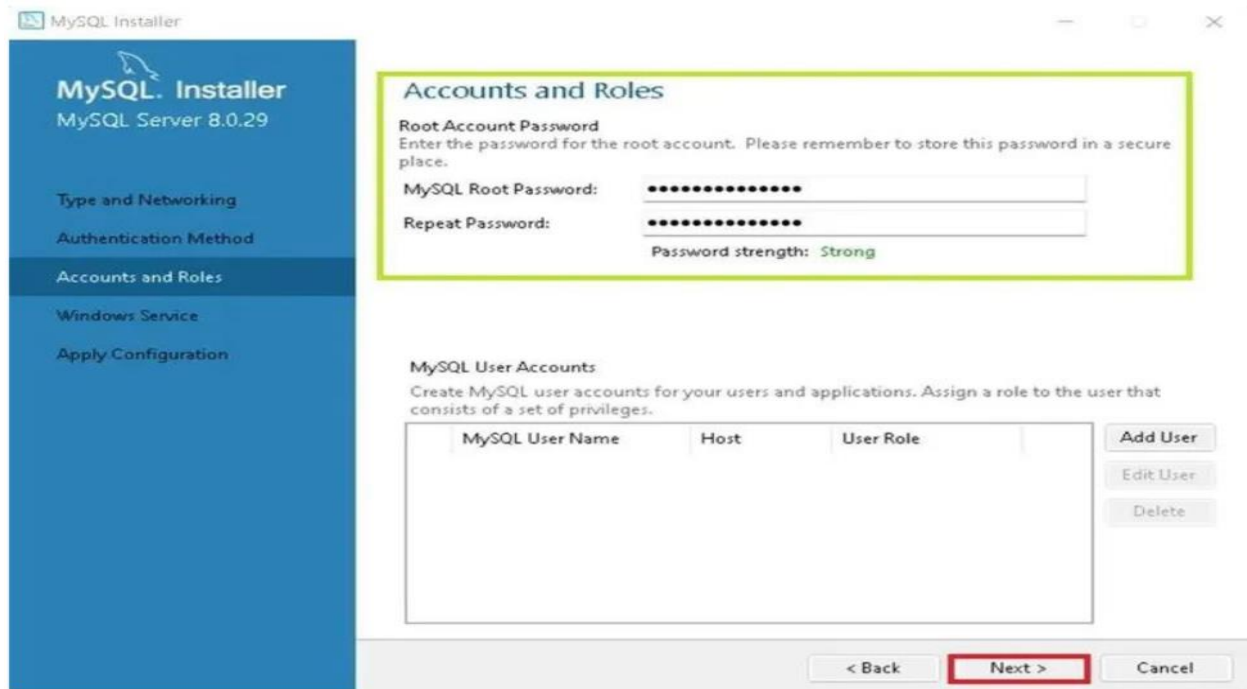
Step 8: Configure the MySQL Server type and networking. Select whether the server will run as a standalone or clustered server, and confirm the default port (3306).



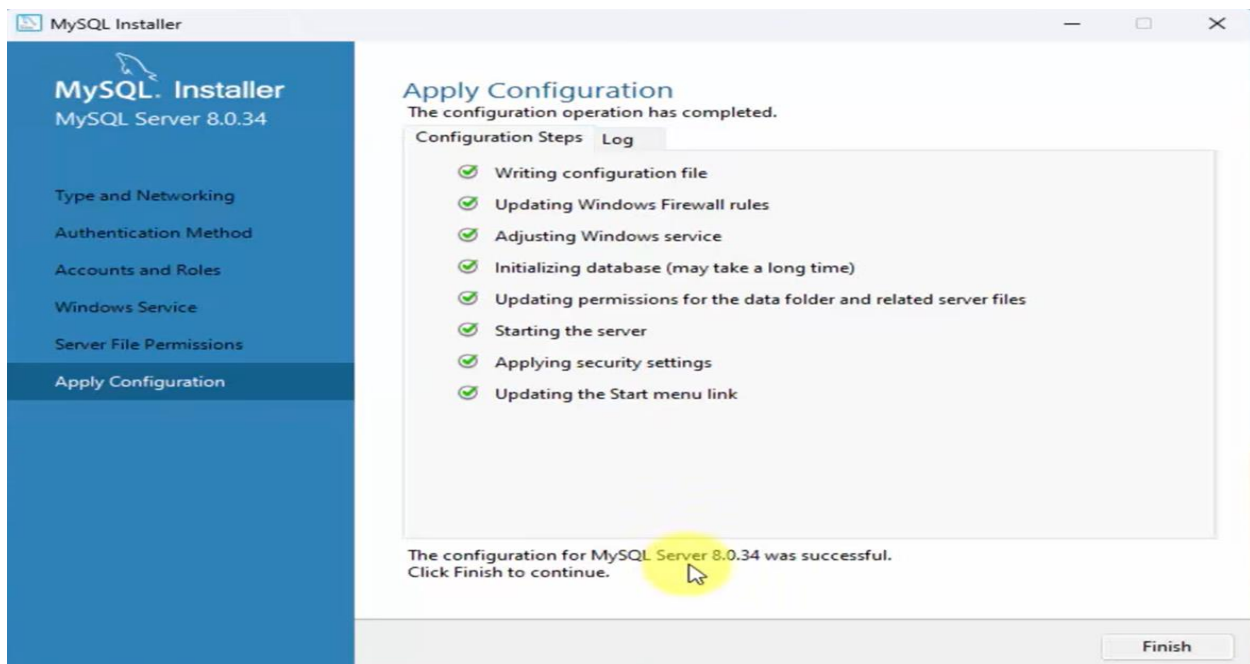
Step 9: Choose the authentication method. Strong password encryption is recommended for better security.



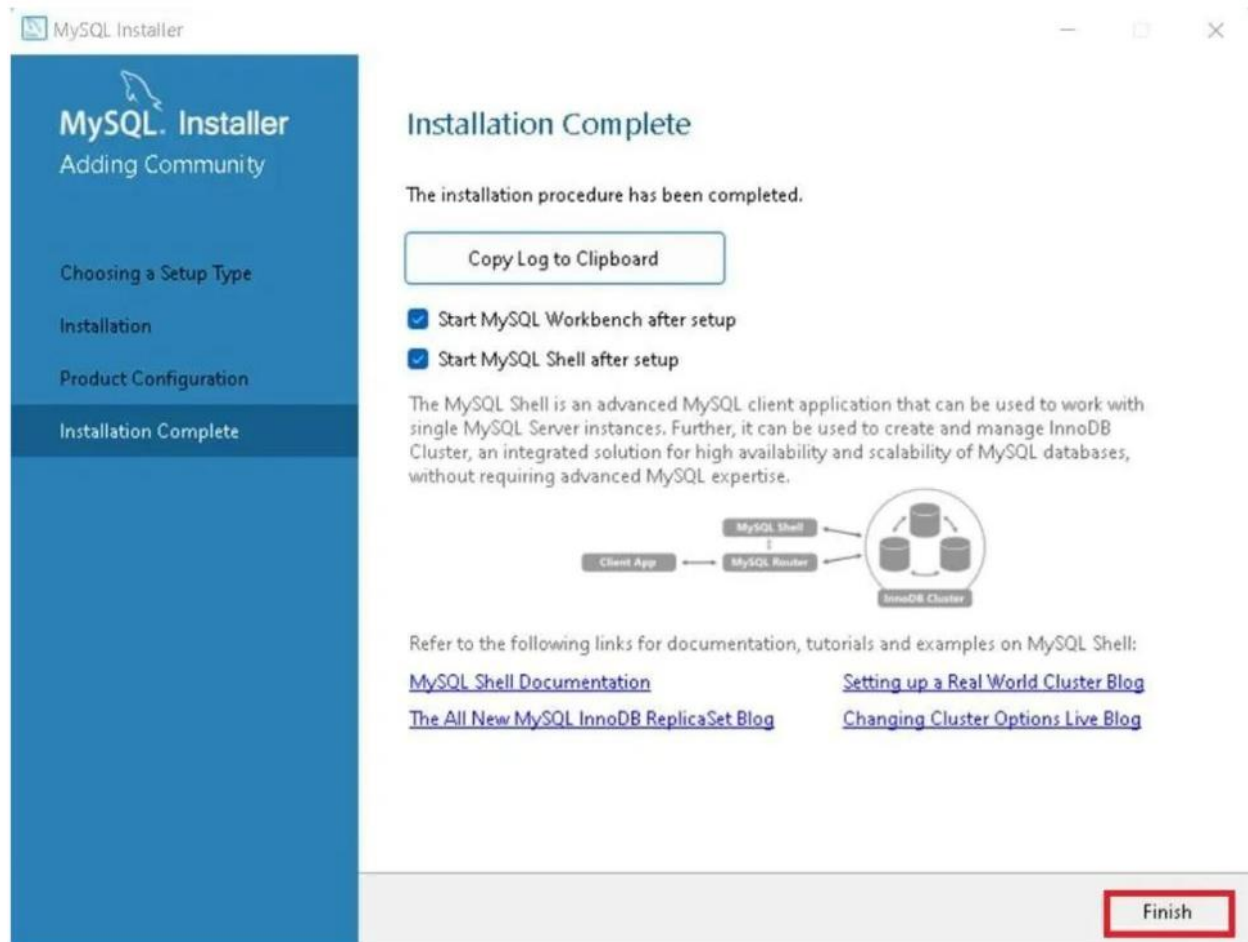
Step 10: Set up the root account by creating a strong password. You may also add additional user accounts and assign roles to them.



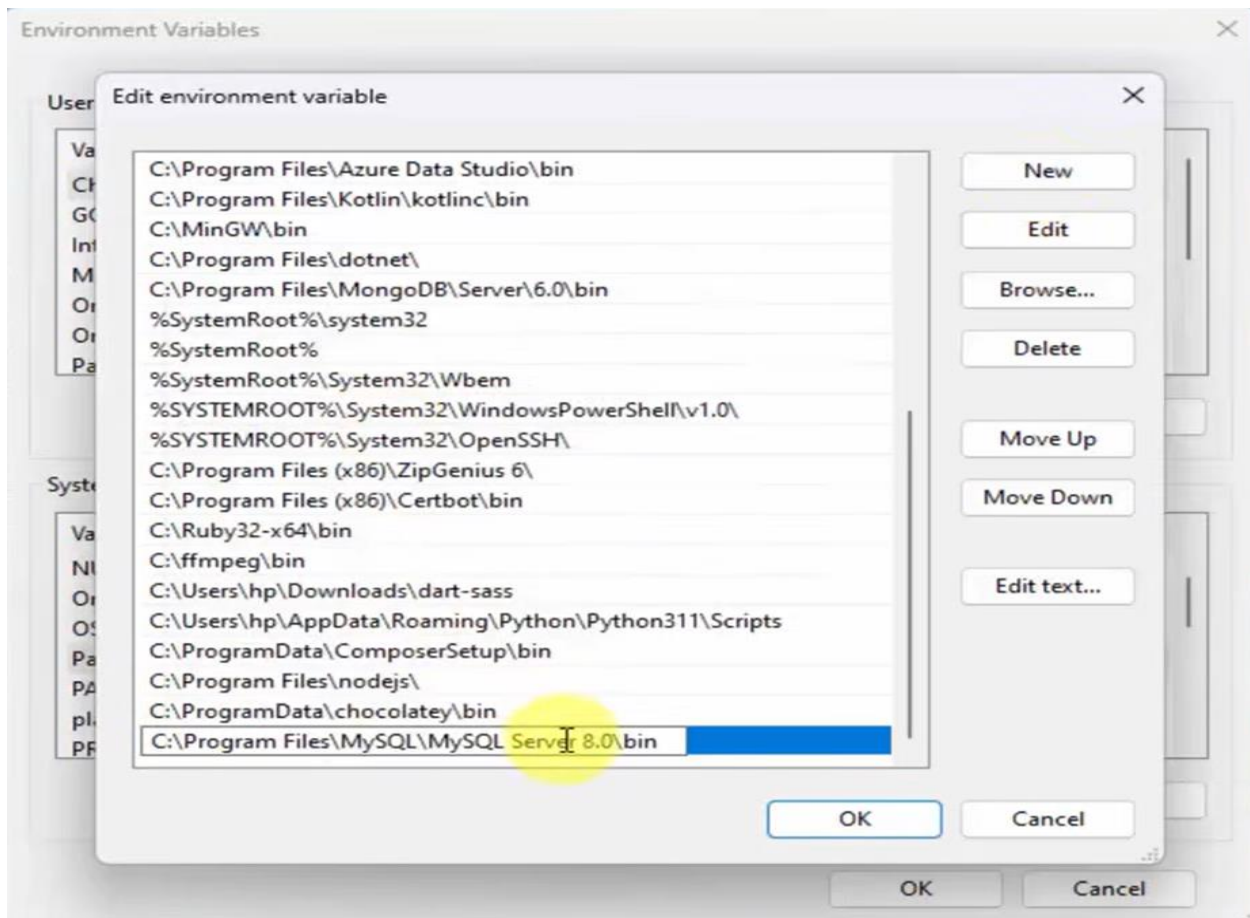
Step 11: Configure MySQL as a Windows Service so that it starts automatically whenever the system boots.



Step 12: Apply the configuration. The installer will test the server connection, finalize the setup, and display a completion screen. You can then launch MySQL Workbench or MySQL Shell to begin using MySQL.



Step 13: Set Environment Variables for MySQL. Add the MySQL bin folder path to the system PATH variable. This allows you to run MySQL commands from Command Prompt without navigating to the installation directory



Step 14: Test MySQL Connection Open Command Prompt, run “mysql -u root -p”, enter your password, and confirm access to the MySQL shell.

```
Command Prompt - mysql -u X + v
Microsoft Windows [Version 10.0.22621.1992]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>mysql --version
mysql Ver 8.0.34 for Win64 on x86_64 (MySQL Community Server - GPL)

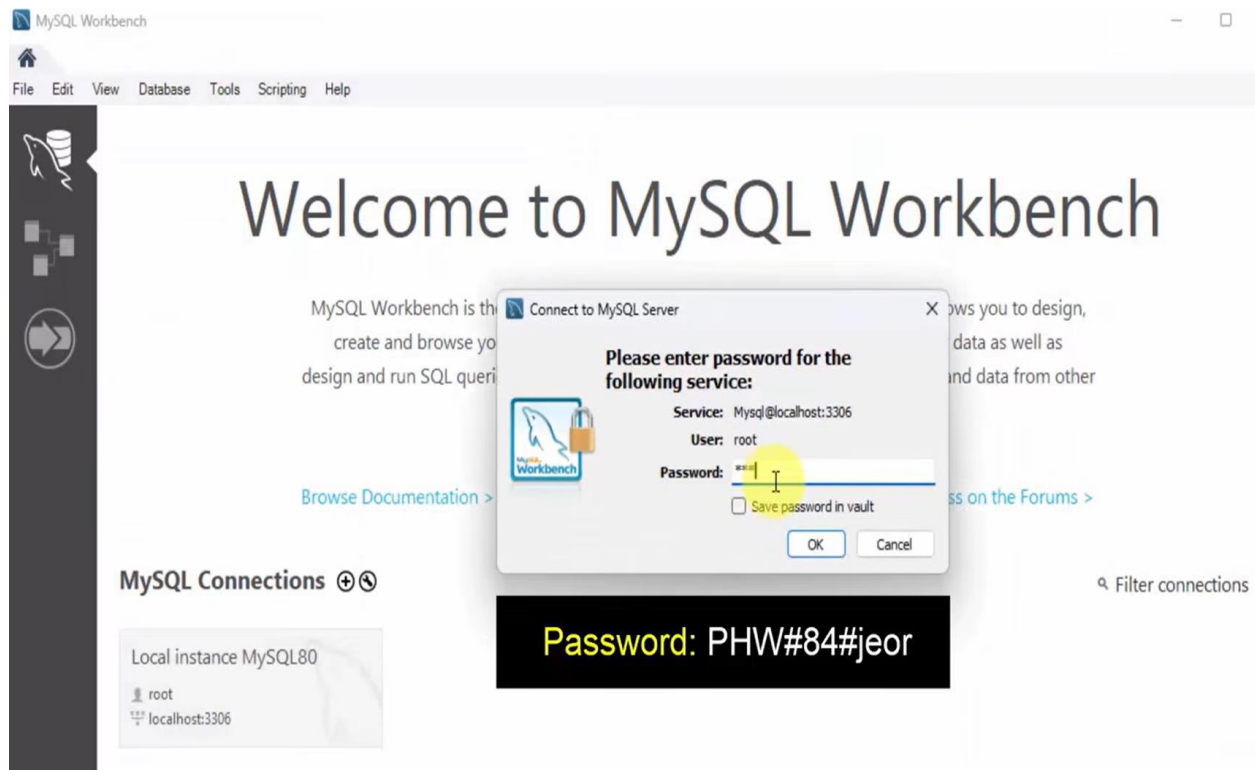
C:\Users\hp>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.34 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> |
```

Step 15: Launch MySQL Workbench Open MySQL Workbench from the Start Menu, connect to the local server using the root account, and begin managing databases through the graphical interface.



Step 16: Ready to Use MySQL With installation, configuration, environment setup, and connection testing complete, MySQL is now fully ready to use for creating databases, running queries, and managing data.